

Maya Elmosalamy

Machine Learning Developer

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Summary

Results-driven **AI/ML Engineer** graduated with **Excellent with Honors** from Misr International University. Experienced in architecting end-to-end machine learning pipelines, deep learning models, and explainable AI (XAI) solutions. **Multilingual** professional proficient in **Arabic** (Native), **English** (Fluent), and **German** (B1).

Education

BS Computer Science, *Misr International University* 09/2022 – 09/2026
GPA: 3.53

Al-Farouk Islamic School, *El-Nozha* 09/2019 – 09/2022

Projects

GeneX — AI-Driven Healthcare Decision Support System *Graduation Project*

Developed a clinical intelligence platform to predict autoimmune disease risks by synthesizing genomic data and healthcare analytics. Integrated **SHAP** and **LIME** to provide clinicians with interpretable model transparency and feature-level risk scoring.

- **System Design:** Architected secure doctor–patient workflows and real-time drug interaction (DDI) alerts.
- **Stack:** Python, Scikit-learn, SHAP, Deep Learning, Healthcare Informatics.

Multilingual Fake News Detection (Arabic & English)

Built a robust NLP framework to classify misinformation across languages, achieving a **93% accuracy**. Evaluated performance across traditional models and advanced architectures including **BiLSTM**, **CNNs**, and **BERT**.

- **Feature Engineering:** Optimized text representation using TF-IDF and word embeddings for linguistic nuances.
- **Stack:** NLP, PyTorch, BERT, BiLSTM, Text Classification, Deep Learning.

Autonomous Lane Detection System

Implemented a real-time lane detection system utilizing semantic segmentation on the **TuSimple dataset**, reaching **95% accuracy**. Evaluated high-efficiency models like **U-Net**, **FastSCNN**, and **ENet** for pixel-level extraction.

- **Optimization:** Leveraged PyTorch for training and used advanced preprocessing to handle varying light conditions.
- **Stack:** PyTorch, Computer Vision, Semantic Segmentation, CNN, Image Processing.

Credit Risk Assessment AI

Designed an intelligent scoring system to evaluate creditworthiness. Utilized **Genetic Algorithms (GA)** for automated feature selection, significantly reducing dimensionality while maintaining model performance.

- **Performance:** Identified a Decision Tree classifier via **GridSearchCV** as the optimal model (75.84% accuracy).
- **Stack:** Scikit-learn, Genetic Algorithms, GridSearchCV, Predictive Analytics.

Experience

Summer Academy Intern, *Dell Technologies* 08/2025

Completed 160+ hours of training in Agile, SWE and AI. Designed a capstone solution to combat online misinformation from both business and technical perspectives. Gained industry exposure via professional shadowing and an on-premise data center tour.

Certificates

- Artificial Intelligence Fundamentals Program - *IBM SkillsBuild*
- Network Security Course - *Cisco*
- Python Programming Basics - *Mahara-Tech*

Skills

Machine Learning & AI

Proficient in PyTorch, TensorFlow, Scikit-learn, and Transformers. Experienced in classical ML models, Deep Learning architectures (CNNs, LSTMs), and Explainable AI (SHAP, LIME).

Programming & Software Development

Proficiency in Python; experienced in JavaScript, C++, Java, C#, and PHP. Skilled in Full-stack development using Flutter and Django.

Data Science & Predictive Modeling

Expertise in feature engineering, genetic algorithms for feature selection, dimensionality reduction (PCA), hyperparameter optimization (GridSearchCV, PSO), and predictive analytics.

Natural Language Processing (NLP)

Strong background in text preprocessing, TF-IDF, multilingual NLP (Arabic/English), and transformer-based text classification models.

Database Management & Tools

Knowledge of MySQL, relational database design, query optimization, and Git/GitHub for version control.

Leadership & Professional Strengths

Proven leadership as ACPC-MIU Media Head; strong skills in technical documentation, presentation, and cross-functional teamwork.

Extracurriculars

Admissions Team Member , <i>MIU Admissions</i>	06/2025
Coding Contestant , <i>Egyptian Collegiate Programming Contest (ECPC)</i>	08/2025
Volunteer , <i>RiseUp Summit</i>	05/2024
Media Head , <i>ACPC Club MIU</i>	07/2024
Managed end-to-end media production, including event photography, video content, and social media strategy. Led recruitment interviews and supervised the media team.	
Media Head , <i>Tuners Club MIU</i>	07/2023
Usher , <i>MIU events & conferences</i>	08/2023
Volunteer , <i>Various Charities</i>	03/2019

Languages

Arabic

English

German
B1